

Code-Layout, Readability and Reusability

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces in Python, 2 in HTML/JS/CSS
2	Proper File Structure	Files and folders are logically organized	Yes	main.py routes, gemini_utils.py AI service, auth.py, models.py, database.py, config.py, templates/, static/, tests/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. remaining_budget, shopping_links, blacklisted_tokens
4	Function / Method Names	Functions are descriptively named	Yes	e.g. get_home_recommendations, build_shopping_links, save_upload_file
5	Code Comments	Inline and block comments explain logic	Yes	Module docstrings and comments for non-obvious logic
6	Modular Design	Code is split into reusable functions/modules	Yes	One module per concern
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared finalize/trim/link helpers serve all three planners; one JS renderer for results and history
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Validation errors (422), auth (401), uploads (413/415), AI failures fall back to default plans

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	build_shopping_links() and PLATFORM_URLS	Python	Home, Party and Jewelry planners, venue suggestions	High
2	_finalize_breakdown() / _trim_to_budget()	Python	Home, Party and Jewelry validation	High
3	_generate_json() Gemini client with model chain	Python	All three planners (text and image)	High
4	renderPlan() JS renderer	JavaScript	Planner result pages and history detail modal	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
5	base.html layout	Jinja2	All 8 pages	High
6	auth.get_current_active_user dependency	Python / FastAPI	Every protected route	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of routes, services, models and storage
Readability	4	Descriptive names and short functions; <code>gemini_utils.py</code> is the longest file
Reusability	5	Shared helpers across planners and pages
Documentation / Comments	4	Docstrings, README and phase documents
Overall Score	4.5	